Module 8, Lesson 2

AI 101 for Mechanical Engineers

Read metrics by their question

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Lesson Structure

1. Core Classification Metrics
2. Applied Evaluation
3. System-Level Ranking

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1. Core Classification Metrics

Define fundamental mathematical metrics for AI classification performance.

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Primary Classification Metrics

Metric	Formula	Evaluative Question
Accuracy	(TP + TN) / N	Correct classification fraction?
Precision	TP / (TP + FP)	Alert validity fraction?
Recall	TP / (TP + FN)	Actual fault detection fraction?

Output values are dimensionless fractions or [%]. Variables denote sample counts.

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The F1 Score Definition

$$F1 = \frac{2TP}{2TP + FP + FN}$$

- Harmonic mean of precision and recall parameters.
- Requires previously defined precision and recall values.
- Yields a dimensionless fraction or percentage [%].
- Not considered a universal mechanical safety objective.

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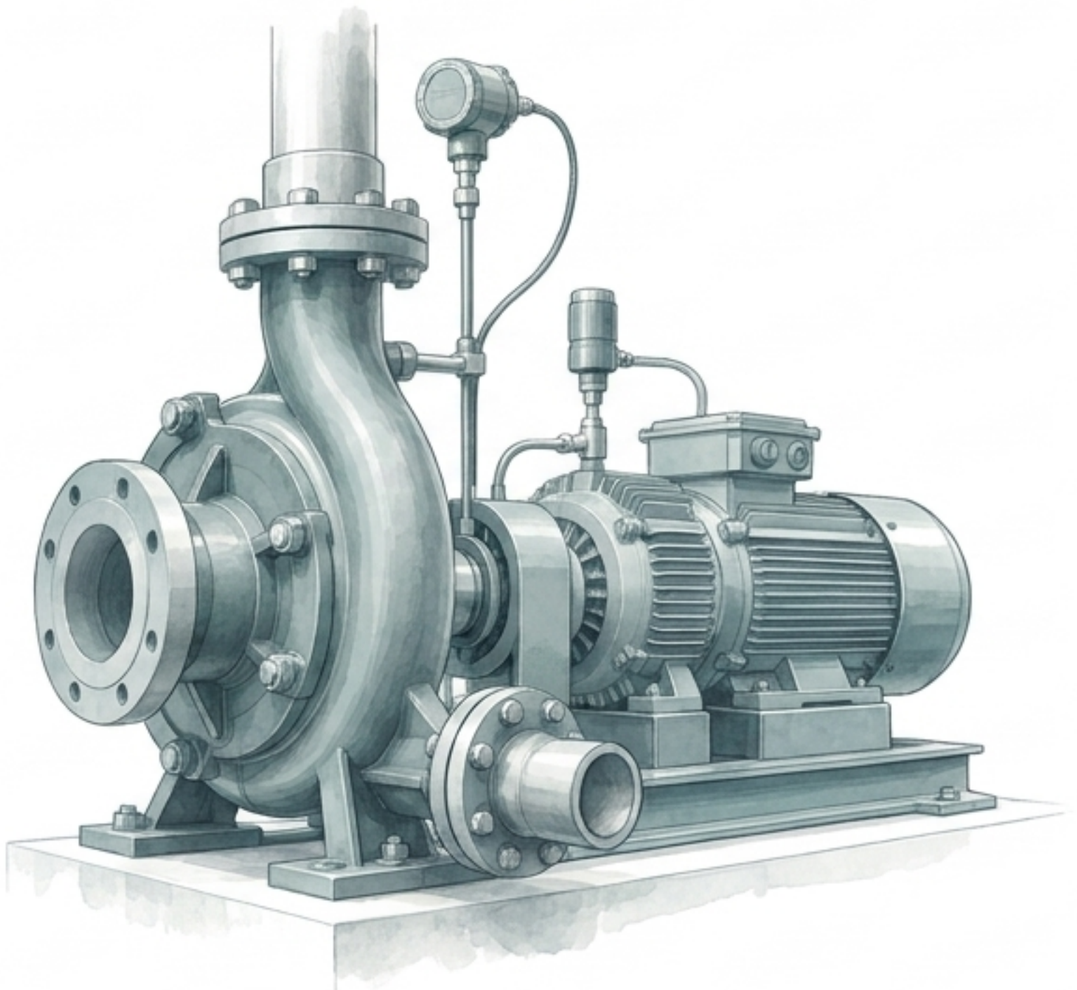


2. Applied Evaluation

Calculate and interpret metrics using a mechanical fault-detection scenario.

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Fault Detection Example Data



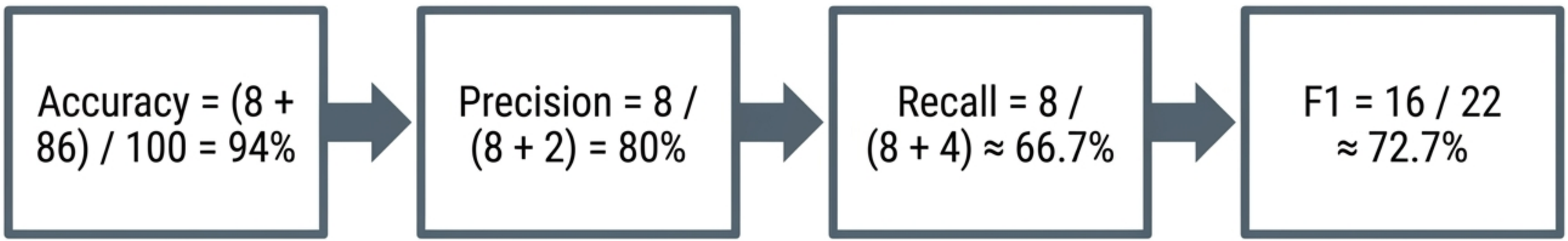
True Positive (TP) = 8	False Positive (FP) = 2
False Negative (FN) = 4	True Negative (TN) = 86

Total evaluated mechanical examples (N) = 100.

Positive class strictly defined as ‘Fault’.

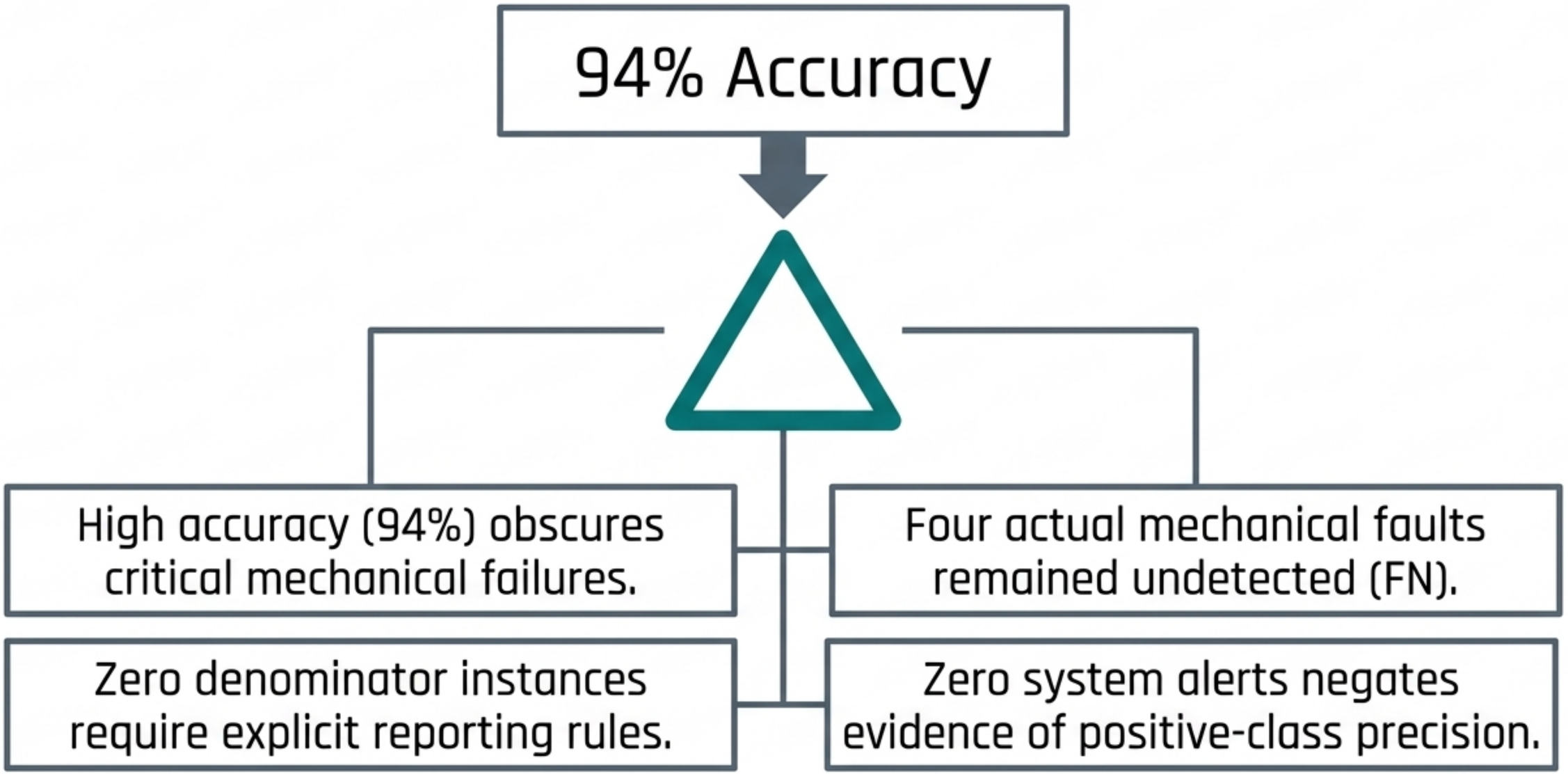
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Calculated Performance Metrics



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Interpreting Missed Faults



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3. System-Level Ranking

Evaluate ranking behaviors and rare fault detection across operational domains.

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ROC-AUC Evaluation



Summarizes ranking behavior across multiple operational thresholds.

Measures discrimination capability through a ROC curve.

Outputs as a dimensionless fractional measure [%].

Requires concurrent checking of relevant operating metrics.

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Domain-Specific Operational Evidence



RULE 1:	Inspect precision-recall performance for rare faults.
RULE 2:	Review specific diagnostic class counts carefully.
RULE 3:	Use validation data to select appropriate thresholds.
RULE 4:	Single numbers cannot replace operating-domain evidence.
RULE 5:	Final evaluation must explicitly report fault consequences.

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Key Points

- Section 1: Calculate core classification metrics to answer specific diagnostic questions.
- Section 2: High accuracy metrics frequently mask critical missed mechanical faults.
- Section 3: Supplement ROC-AUC ranking metrics with operating-domain evidence and class counts.

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